Carl H. Lindner College of Business
University of Cincinnati

Financial Machine Learning and AI

FIN 7057 (3 credit hours)
Fall 2026

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Logistics

- **Instructor:** Shane Conway
- **Office:** Remote
- **Email:** conwayso@uc.edu
- **Course website:**
  - Canvas: http://uc.instructure.com/
  - GitHub: https://github.com/smc77/uc_finmlai
- **Office hours:** Two hours each week, split across two days (times TBD): open drop-in on Zoom, plus bookable slots through the Canvas scheduler — 10 minutes for a quick question, 20 minutes for project or debugging help.
- **Communication policy:** Post course questions to the Canvas discussion board, where the whole class benefits from the answer; email me for anything personal. I reply within one business day, weekends excepted. For help working through the material, please use office hours or the discussion board rather than one-off calls. Questions posted less than 24 hours before a deadline may not get a reply before it is due — plan accordingly.

Course Description

This course is an integrated introduction to the research workflow, the major modeling approaches, and the evaluation practices of machine learning and artificial intelligence in finance. Students learn fundamental ML algorithms and their application to financial problems including return prediction, risk management, portfolio optimization, derivative pricing, and corporate finance. Advanced topics include large language models for financial text analysis, market microstructure and optimal execution, reinforcement learning for sequential decisions, and causal inference for moving from prediction to intervention. The course closes with the governance questions — fairness, model risk, and regulation — that deployed models must address. The course emphasizes both theoretical foundations and practical implementation in Python, and the unique challenges of applying AI in financial markets. This is a course in financial ML **research** — the workflow for producing credible, decision-relevant results — rather than production ML engineering; deployment, monitoring, and governance are covered at the depth a researcher must understand, not built as infrastructure.

Course Philosophy

This course teaches financial machine learning as the algorithms together with the **research workflow** that makes them work: turning noisy, non-stationary data into predictions, decisions, and risk-aware evaluations you can defend.

You will learn the modern methods and how to measure them against simple benchmarks, so you can tell which apparent improvements are real. In finance, telling a real edge from a lucky one is one of the most valuable skills the course can teach.

Scope and Depth

The course is deliberately selective about depth. A **core** set of skills — leakage-free data handling, time-aware validation, baselines you must beat, cost- and decision-aware evaluation, and turning predictions into decisions — you are expected to implement and defend. A wider **applied** layer — classification and credit scoring, factor models, neural networks, financial NLP and retrieval, options pricing — you should be able to use and know when to reach for. A **frontier** set — hidden Markov models, deep hedging, reinforcement learning, causal inference, and conformal prediction — is taught for informed exposure: understand what each is for and where it breaks, rather than implement it from scratch. Homework and quizzes concentrate on the core; the frontier topics make you a literate consumer of methods you will meet in practice.

Learning Objectives

Students who complete this course will be able to:

Build reliable financial machine learning systems.

- Prepare financial data, engineer features, and develop predictive models using modern machine-learning methods.
- Judge when different modeling approaches are appropriate, from linear models through neural networks and large language models.

Evaluate models the way practitioners and researchers do.

- Design rigorous out-of-sample evaluations that account for time, transaction costs, overfitting, and other sources of false confidence.
- Compare sophisticated models against appropriate baselines and judge rigorously which improvements are real.

Translate predictions into financial decisions.

- Move beyond prediction to portfolio construction, risk management, execution, and other real financial decision problems.
- Distinguish prediction, optimization, and causal inference.

Apply modern AI responsibly.

- Use generative AI and large language models effectively for financial analysis while understanding their limitations, failure modes, and appropriate evaluation.
- Recognize where reinforcement learning and other advanced AI methods are useful — and where they are not.

Operate as a professional quantitative researcher.

- Design reproducible research workflows, communicate results clearly, and weigh the ethical, regulatory, and governance considerations surrounding AI in finance.

Prerequisites

FIN 7014 and working Python. **Readiness standard:** you should enter able to interpret a linear regression, common probability distributions, and hypothesis tests; work with vectors, matrices, and basic optimization ideas; and write and debug Python using NumPy and pandas. The back half of the course (neural networks, sequence models, reinforcement learning) leans on probability and linear algebra.

Week 0 is one 60-minute setup orientation plus a setup notebook. Its three blocks establish the course repository on GitHub, demonstrate optional accountable use of Claude Code or Codex, and verify a working Colab or local-Python notebook path. It is ungraded but **strongly expected unless you comfortably pass the readiness self-check**, which points you to a Python/NumPy/pandas refresher if needed before Week 1.

Course Materials

The lectures are self-contained. Everything required for the homework and quizzes is taught in the lecture videos and their slides. There is no required reading; you do not need to buy or read anything to complete the course.

Course repository (the one thing you need). All lecture notebooks and homework assignments are released each week in the course GitHub repository. Clone it at the start of the term and pull each week. Every example notebook opens directly in Google Colab with one click, so you can run everything in the browser without local setup. The companion `finmlsim` package installs with `pip install finmlsim`.

Optional weekly podcast — *The AI Quant Weekly.* Each week has a short optional audio episode (about 15–20 minutes) that introduces the core ideas and the tension behind them in a relaxed, conversational format. It is meant as a running start to listen to *before* the week's lecture — on a commute or a walk — if you find it useful. Like the background reading, it is entirely optional and no assessment depends on it; the lectures remain the complete, self-contained path.

Background text (optional, for going deeper): Conway, *Financial Machine Learning and AI: Simulation, Signals, and Evaluation*. Each week's schedule lists the chapters that align with the lecture, offered as further study for students who want more depth, derivations, or additional worked examples. This reading is never required and no assessment depends on it.

Optional supplements:

- Dixon, Halperin & Bilokon, *Machine Learning in Finance: From Theory to Practice* (ISBN 9783030410674) — alternative treatment of the core methods.
- Kelly & Xiu, *Financial Machine Learning* (SSRN) — research / empirical asset-pricing perspective (return-prediction and factor weeks).
- López de Prado, *Advances in Financial Machine Learning* (ISBN 9781119482086) — backtesting, leakage, and validation (used selectively).
- James, Witten, Hastie, Tibshirani & Taylor, *An Introduction to Statistical Learning, with Applications in Python* (free at statlearning.com) — general ML / statistical-learning foundation, not finance-specific; a gentle reference for the core methods.

Selected readings from recent papers and from financial-econometrics sources will be posted to Canvas.

Assessment

Component	Weight
Homework (7)	35%
Quizzes (14)	20%
Final Project	35%
Project milestones	10%
Bonus	up to 3%

Bonus work may add up to three percentage points to your final course grade, capped at 100%. It comes from the **optional advanced extensions** on the homework assignments and an **optional extension of the final project** — a chance to reach past the required core. Bonus work is open to everyone, clearly scoped, truly additional, and graded on the same standard as the rest of the course.

Quizzes

Fourteen short weekly quizzes on the prior week's material, taken in Canvas, all multiple choice so they grade automatically. Open-book and open-note: you may use the text, your own notes, and the course materials, but **not AI assistants or another person.** Using AI on a quiz where it is prohibited puts you at risk of an academic-integrity violation at UC. Just as important, these quizzes exist to confirm that you actually understand the material — leaning on AI to produce the answers only keeps you from learning what you will need to progress in the course.

Homework

Seven programming assignments revisit one research discipline while changing the financial problem when the topic requires it. HW1–3 share a time-series pipeline; HW4–5 move to a cross-sectional panel for factor structure and portfolios; HW6 begins the student's chosen final project; and HW7 is a deliberate text-data transfer exercise. What accumulates is the workflow — availability-aware data, leak-free features, baselines, time-aware selection, decision-relevant evaluation, and reproducibility — rather than an artificial requirement to keep one dataset all term.

Each assignment ships a **self-check cell** of automated tests for the mechanical parts that can be tested reliably. The checks catch common implementation errors but do not certify economic reasoning or the absence of every possible leak; grading also weighs methodological choices, interpretation, and limitations. Once an assessment period has been inspected, it is spent and may not be presented later as untouched evidence. Later assignments therefore use a new forward window, an outer fold, or a reserved lockbox as specified in the assignment. After each assignment a **reference solution** is posted, so an early mistake need not compound. Each assignment has a **required core** and truly additional **optional extensions**; maximal sophistication is never the implicit norm.

HW	Due after	Assignment
HW1	Week 3	Financial data audit & leakage-free feature engineering
HW2	Week 5	Linear models + regularization; in-sample vs. out-of-sample with time-series validation
HW3	Week 7	Trees & boosting, **evaluated out of sample with realistic costs**
HW4	Week 8	Unsupervised mini-lab: PCA/factor structure and/or regimes (short notebook)
HW5	Week 9	Prediction → portfolio: convert scores into a long-short portfolio with turnover and costs
HW6	Weeks 10–11	Final-project proposal & baseline checkpoint
HW7	Week 13	*Short* financial text / LLM application **with an evaluation** (faithfulness and leakage checks) — lighter than HW1–6, to protect final-project time

Each assignment falls in the week after the chapters it rests on. HW3 follows the evaluation chapters (Ch 15–16, Week 6), so boosting is judged out of sample with costs rather than on raw accuracy; HW5 immediately follows the portfolio-construction week (Ch 22–23, Week 8).

AI may assist with coding, analysis, and revision as described in the AI-Use Policy, but your submitted reasoning must reflect judgments you understand, verify, and can defend.

Final Project

Develop an end-to-end machine-learning solution to a financial problem. Choose a track:

Track	Example problems
Markets / asset management	Return prediction, volatility forecasting, factor timing, long-short portfolios
Corporate / credit / text	Credit default, bankruptcy prediction, filing analysis, earnings-call sentiment, risk summaries

Every project, regardless of track, must include:

1. **Research question** — what financial question are you answering?
2. **Data pipeline** — what data, and *when would it actually have been available*?
3. **Feature engineering** — what predictors, and why?
4. **Baseline model** — a simple linear, historical, or otherwise naive benchmark to beat.

5. **ML model** — at least one method meaningfully more flexible than the baseline and appropriate to your data and sample size (regularized regression, gradient boosting, or a neural network — chosen for fit, not novelty).
6. **Out-of-sample evaluation** — time-aware validation, with the right metric for your track.
7. **Decision layer** — translate predictions into a portfolio, classification decision, risk flag, or action.
8. **Limitations** — what would make this fail in reality?
9. **Pre-analysis plan** — a short protocol (hypothesis, data and clock, procedure, and the pass-or-fail bar) committed at the proposal stage, before the test set is opened. It mirrors the pre-registered investigations in the book: commit to what would count as success before you can see whether you got it.

Negative results are not penalized. A well-executed project that finds *no* robust predictive signal can receive full marks if the research design, evaluation, and interpretation are strong. Well-reasoned, self-critical skepticism is rewarded over a result that "works."

Reproducibility (required). The project must be reproducible from the submitted code and must document data sources, sample periods, train/test splits, fixed random seeds, the environment/package list, and any AI tools used.

Data and computing. Projects must be reproducible using course-accessible data and reasonable compute. Any paid, restricted, or non-redistributable source (Bloomberg, WRDS, CRSP/Compustat, employer data) needs prior approval and a **functionally reproducible** alternative: a redistributable derived dataset where licensing permits, a course-accessible substitute, or enough synthetic or sample data to reproduce and evaluate the full pipeline. Documented access instructions alone do not make a project reproducible for someone who lacks the access. The core pipeline should run in under roughly 30–60 minutes on Colab or a standard laptop (optional extensions excepted).

Milestones (checkpoints graded 10%, on completion).

- *Week 6:* one-paragraph project idea (so data-availability problems surface early).
- *Weeks 10–11:* full proposal + baseline checkpoint (HW6), including the pre-analysis plan committed before the test set is opened.
- *Week 15:* final presentation + report with code.

HW6 is graded as a homework assignment for the quality of the proposal and baseline analysis; the milestone grade separately rewards timely completion of these project checkpoints.

Format. Individual. Deliverables: a ~10–12 minute presentation (12–16 slides), a reproducible code repository or notebook, and a 6–10 page report; see the project brief for specifics.

Citations. Use any recognized citation style consistently, with persistent links or DOIs where available. Cite datasets, software packages, pretrained models, and LLM providers, not only papers.

Grading rubric.

Component	Weight
Research question & economic rationale	10%
Data provenance & point-in-time integrity	15%
Baseline & model design	15%

Component	Weight
Validation & robustness	20%
Decision relevance & economic/business interpretation	15%
Interpretation & limitations	15%
Reproducibility & communication	10%

Metrics by track. Use the metrics that fit your decision, not the markets defaults by reflex:

	Markets / asset management	Corporate / credit / text
Prediction	return / rank prediction	probability / classification
Evaluation	IC, Sharpe, drawdown	AUROC, PR-AUC, log loss, **calibration**
Costs	turnover & transaction costs	decision thresholds & asymmetric costs
Decision layer	portfolio	lending / risk-review / escalation
Validation	walk-forward	temporal / cohort
Availability	market-data timing	filing / publication / accounting timing

AI-Use Policy

AI tools may assist any stage of your work unless an assignment says otherwise. But **you remain responsible for everything you submit** — every claim, decision, line of code, citation, and interpretation — and you must be able to explain and defend it without the tool. Undisclosed or unverified AI output is treated as your own work and graded as such: if it is wrong, the error is yours. From that principle: **AI use is prohibited on quizzes**; any material use elsewhere must be disclosed (below); submitted code must be understood; data and citation claims must be independently verified; and you may be asked to **explain any submission** in person. A citation counts as verified only when you have found the original source, confirmed it exists, and checked that it supports the claim you cite it for — for empirical claims, note the relevant page, table, or passage.

Disclosure. When AI materially assists an assignment, include a short note: the tool or model, what you used it for, the material prompts or workflow, what output you incorporated, how you checked it, and what stayed your own judgment. Incidental autocomplete need not be logged; material assistance — and any use of AI as the object of study — must be.

Data confidentiality. Do not upload confidential, employer-owned, personally identifiable, restricted, or license-protected data or code to an AI service or any external tool. Use only course-approved public or synthetic data unless you have explicit written permission. Handling client and firm data with care is part of the profession this course trains you for.

In HW7, students may choose a classical text model, an embedding-based system, or an LLM application. When an LLM is the chosen method or object of evaluation, using the model is required and students must document model choice, prompts or system instructions, evaluation criteria, and known limitations. The ordinary disclosure and verification requirements apply to every track.

Reproducibility & Research Hygiene

All submitted work should use recorded random seeds, state train/test dates explicitly, include an environment file or package list, document data sources, and be organized as clean, runnable notebooks or scripts. Where stochastic variation could materially affect a result — neural networks, clustering, HMMs, reinforcement learning — report sensitivity across several seeds rather than a single lucky one. These practices are graded as part of homework and the final project.

Late Work, Extensions & Collaboration

Late homework. You have one **48-hour grace token** to spend on any single assignment, no explanation needed. Beyond that, late work is penalized 10% per day up to three days, after which it is not accepted. Documented emergencies (illness, family, university-recognized) are always accommodated — reach out as early as you can.

Quizzes. Your lowest quiz score is dropped; after the drop, the remaining thirteen quizzes are weighted equally. There are no make-up quizzes except for documented emergencies.

Project milestones (10%). The milestone checkpoints are lightly graded on completion — a feasible question, a point-in-time data check, and a working baseline — to reward getting the project actually underway. They still receive formative feedback, and the final deliverable is firm.

Technical problems. A technical problem does not by itself extend a deadline, but documented platform-wide outages or failures of required course infrastructure will be accommodated. Commit and back up your work often, and do not begin an upload right at the deadline.

Collaboration. You are encouraged to discuss concepts, readings, and general debugging strategies with classmates. The code and written analysis you submit must be your own — do not share or copy notebooks or solution code. Homework and the final project are individual. When in doubt, ask.

Grading Scale

A	93.33–100.0	C	73.33–76.65
A-	90.00–93.32	C-	70.00–73.32
B+	86.66–89.99	D+	66.66–69.99
B	83.33–86.65	D	63.33–66.65
B-	80.00–83.32	D-	60.00–63.32
C+	76.66–79.99	F	00.00–59.99

Course Schedule

Each week has **two 60-minute lecture parts** (A and B), each split into several recorded sections with individual work between them — a Jupyter notebook to run or a short mini-project — for about **two hours of recorded lecture and one hour of hands-on work**. The **Reading** column lists the textbook chapters that align with each week as optional background; the lectures cover everything the course requires.

Expected weekly effort (an average — it varies): roughly 2 hours of recorded lecture and 1 hour of interspersed hands-on work, optional background reading, and a short quiz, plus **typically 2+ hours** of programming on homework weeks (more if you are still building Python fluency).

Wk	Part	Topic	Reading	Application	Due
0	—	Setup & onboarding *(one hour; ungraded; expected unless the readiness self-check is passed)*	GitHub, Claude/Codex, Colab/Python	Repository access, optional agent review loop, clean notebook run; readiness self-check	—
1	I	The financial-ML mindset, research workflow, time series & stylized facts	Ch 1–4	Reproducible research workflow: notebook structure, data provenance, experiment configuration, AI-assisted coding and verification; prices vs. returns, stationarity; fat tails and volatility clustering	—
2	I	Features, transformations & the leakage taxonomy	Ch 5–6	Leakage-free momentum/volatility features and target alignment; scaling, distribution-shaping and feature filtering (sklearn transforms under a point-in-time boundary); why shuffling breaks time-series validation; survivorship & delisting bias, restatements and revised data	Q1
3	II	Regression models & forecast evaluation	Ch 7, 9	OLS as the load-bearing baseline, then subset/stepwise selection, ridge/lasso/elastic-net shrinkage, and robust/quantile regression (ESL Ch 3); time-series Spearman correlation and rank IC; trace forecast → position → return → cost in a first long-short	HW1, Q2
4	II	Classification, calibration & validation	Ch 8, 10	Credit/default classification and probability calibration (reliability, Brier, log loss, thresholds); walk-forward and purged/embargoed CV across three settings — time series, cross-sectional/panel, event/document; model selection	Q3
5	II	Trees, forests, boosting & tuning	Ch 11–14	Credit scoring, cross-sectional return prediction with XGBoost/LightGBM; time-aware early stopping; feature importance is not a causal mechanism	HW2, Q4

Wk	Part	Topic	Reading	Application	Due
6	II	Evaluation at scale, regimes & when complexity helps	Ch 15–18, 36	Rank IC and the fundamental law (and its assumptions); costs, turnover, multiple testing, deflated Sharpe, and the informal search a clean test set can't catch (researcher degrees of freedom); regime switching (HMMs as a latent-state concept); is complexity genuine learning or recency-harvesting?; prediction intervals and conformal coverage as an uncertainty extension (Ch 36)	Q5 *(project idea due)*
7	III	Optimization, PCA & clustering	Ch 19–21	Loss functions to trading decisions; PCA and factor structure, clustering; regime instability and the tradeability caveat	HW3, Q6
8	III→IV	Portfolios & deep-learning foundations	Ch 22–25	Sizing, turnover, vol targeting, mean-variance fragility; neural networks, function approximation, sequences	HW4, Q7
9	IV	Case study: options pricing & deep hedging	Ch 26–27	Learning a pricing *function* vs learning a hedging *policy*; neural option pricers and vol surfaces; hedging under frictions	HW5, Q8
10	IV	Text as data & sentiment	Ch 28–29	Bag-of-words to transformers, the finance lexicon; earnings calls, SEC filings, timestamp leakage	Q9
11	IV	LLMs, retrieval & evaluation	Ch 30–31	Retrieval-augmented generation over filings; faithfulness, hallucination, prompt injection	HW6, Q10
12	V	Market microstructure & reinforcement-learning foundations	Ch 32–33	The limit order book and optimal execution; MDPs and Q-learning	Q11
13	V	Two kinds of decision: policies vs. intervention effects	Ch 34–35	*Lecture A — learning a policy:* RL, states/actions/rewards, optimal execution, off-policy pitfalls; *Lecture B — learning an intervention effect:* potential outcomes, identification vs. prediction, difference-in-differences worked in depth (IV/RDD/uplift as a closing map)	HW7, Q12

Wk	Part	Topic	Reading	Application	Due
14	VI	From research result to a governed model	Ch 37–38	*Lecture A:* fairness and adverse impact (proxy discrimination, the 4/5ths screening heuristic), the SR 11-7 to SR 26-2 model-risk transition (risk-based and proportionate — and notably leaving generative and agentic AI outside its formal scope, so supervisory frameworks visibly lag technological practice), reason codes, monitoring and drift, human oversight, model retirement; *Lecture B:* final-project clinic & synthesis	Q13
15	—	Capstone presentations & course synthesis	Rules of the Craft appendix	Final presentations; the recurring themes, in review	Final; Q14

Timing may shift as the term progresses; any changes will be announced on Canvas.

Academic Integrity

This course upholds the highest ethical standards. LCB instructors are required to report any incident of academic misconduct (e.g., cheating, plagiarism) to the college review process, which can carry severe consequences including dismissal. See the UC Code of Conduct.

Accessibility Resources

Students with disabilities have the right to full and equal access at the University of Cincinnati. The Accessibility Resources office will work with you and your instructors to identify reasonable accommodations. Contact: AccessResources@uc.edu.

Religious Accommodations

Ohio law and the University's Student Religious Accommodations for Courses Policy 1.3.7 permit students, upon request, to be absent or receive alternative accommodations for reasons of faith or religious/spiritual belief. Provide written notice of specific dates within fourteen days of the first day of instruction. For more information contact the Office of Equal Opportunity and Access: (513) 556-5503, oeohelp@ucmail.uc.edu.

Title IX

The University of Cincinnati is committed to a learning environment free from sex- and gender-based discrimination, harassment, and violence. Title IX resources and reporting options are available through the UC Office of Gender Equity & Inclusion.

Mental Health & Well-Being

If you are struggling — academically or personally — please reach out. UC Counseling & Psychological Services (CAPS) offers confidential support.

Class Recording

Recorded lectures and course materials are provided for enrolled students' educational use only and may not be redistributed or posted publicly without permission.